

"PAPER_{0:D3}" -f [int]# PAPER #76 ■ Gamma-Ray Sources: Fermi-LAT + UQFF Emission Model

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Where:

$A_R = 10^{-5}$ (Resonant mode amplitude)
 $\omega_{\text{blazar}} = \text{system spin/jet frequency (typically } 10^{-7} \text{ to } 10^{-5} \text{ rad/s for BL Lac objects)}$

For Markarian 421 (BL Lac, d=134 Mpc):

$\omega_{\text{Mrk421}} = 2\pi/(315 \text{ days}) = 2.31 \times 10^{-7} \text{ rad/s}$
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$$m_{\gamma, \text{UQFF}}^2 = \frac{\hbar^2 \times \rho_{\text{vac, [UA]}} \times c^2}{\epsilon_0} = 1.05 \times 10^{-70} \text{ kg}^2$$

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The Fermi-LAT 4FGL catalog lists the Crab as **4FGL J0534.5+2201** with flux (100 MeV-100 GeV) = 5.65-10⁻⁷ ph/cm²/s. UQFF does not modify the average flux but predicts a 10⁻⁵ amplitude modulation component - below 4FGL timing precision for single-pulse analysis but potentially detectable in epoch-folded analysis.

4. 4FGL Source Comparison Table

4FGL Source	Type	UQFF Prediction	Fermi-LAT Constraint
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J1653.9-0158	BL Lac (ultrafast)	Resonant peak	Not constrained

Summary

Gamma-Ray Observable	Standard GR+QED	UQFF Prediction	4FGL Status
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Spectral shape	Power law/cutoff	Unmodified ($m_\gamma \sim 10^{-5}$ eV)	Match
Flux periodicity	Source-dependent	+10% modulation	Below sensitivity
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Source: `QCalcvalidation.py FERMILAT + FERMI4FGL endpoints | ? = 0.0005/day | [SSq] = 0.57`
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For Markarian 421 (BL Lac, $d=134$ Mpc):

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Summary

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Summary

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Average flux	Eddington-based	Unmodified	Match
Spectral shape	Power law/cutoff	Unmodified (m_? ~ 10?5 eV)	Match
Flux periodicity	Source-dependent	+10?5 modulation	Below sensitivity
Phase-pulse profile	Fixed	+10?5 at peak phase	Not constrained

Source: QCalcvalidation.py FERMILAT + FERMI_4FGL endpoints | ? = 0.0005/day | [SSq] = 0.57

UQFF computed: Canonical UQFF buoyancy parameter $U_{bi} = \frac{[SSq]GM}{r} = 5.0e-4 \frac{0.57 \cdot 6.67e-11 M}{r}$; for solar parameters: $U_{bi,Sun} = 5.7e-4 \frac{6.67e-11 \cdot 1.99e30}{(6.96e8)} = 1.47e+2 \text{ m/s}$.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-11} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{22} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{\{g1\}} + U_{\{g2\}} + U_{\{g3\}} + U_{\{g4\}} + U_{\{bi\}} + U_m - \sum_{i=1}^4 i g_l[\lambda_i \cdot U_i(r,t) \cdot E_{\text{react}}] i g_r]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\Sigma \lambda_{\text{U}} \cdot \mathbf{U} \cdot \mathbf{E}_{\text{react}}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420): $\lambda_{\text{U}}=10^{-10}$, $\lambda_{\text{U}}=10^{-12}$, $\lambda_{\text{U}}=10^{-11}$, $\lambda_{\text{U}}=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_{\text{m}}^{\text{full}} = U_{\text{m}}^{\text{base}} \times \text{sigl}(1 + 10^{13} \cdot \Theta(\text{ho}_{\text{SCm}} - \text{ho}_{\text{c}})) \times \text{sigl}(1 + A_{\text{q}} \cos(\Delta\omega t))$$

Symbol	Value	Description
ρ_{c}	10^1 kg/m^3	SCm critical superconducting density
A_{q}	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_{\text{i}} \times \text{Ubi}$	Expanding nebulae, stellar winds
Superconductive	$\text{Um} \times (1 + 10^{13} \cdot f_{\text{H}})$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.